Continued fractions

Presenter notes for two 50-minute lectures

The slides follow Chapter 3 of *Patterns, Proofs, and Primes*. Each lecture has its own 50-minute route. The suggested times include student calculation and discussion. Answer slides follow question slides, so advancing the PDF provides a simple reveal without animation software. The optional slides come after the exit question.

Before the first lecture. Students should know how to multiply and invert a 2×2 matrix, how a simple continued fraction is written, and what its convergents are. Lecture 1 reviews the transformation rules. For Lecture 2, recall that the convergents tend to the original irrational number and their denominators are eventually unbounded. No calculus or abstract algebra is needed for these proofs.

Using the files. Share only the slide PDF window in Zoom and keep this notes document separate. If you have one screen, practice switching to the notes without changing the shared window. The editable LaTeX sources are included in the source bundle. No live web services or fonts need to download during the lectures.

Pacing. Lecture 1 reaches the prefix-conjugation identity at minute 29 and finishes that argument by minute 35. Lecture 2 completes Lagrange's converse at minute 24, begins the Galois discussion at minute 30, and completes its proof at minute 41. If discussion runs long, shorten the final comparison examples. Keep the student calculation breaks whenever possible.

Lecture 1: Continued fractions and matrix fixed points

1. Continued fractions and matrix fixed points

Minutes 0 to 1

Open with the question: why should repeating continued fractions know anything about quadratic equations? Students have already met linear fractional transformations. Treat the first few slides as a review, not a new chapter of prerequisites. Today's destination is the prefix-conjugation argument. Source: course book, Sections 3.2 and 3.7.

2. A repeating pattern

Minutes 1 to 3

Ask students which digit belongs to the repeating block in each expression. Explain that pure periodicity includes the integer part. A bar over the block means to repeat it indefinitely. The original number and its complete quotient can have different kinds of periodicity. Recall that rational simple continued fractions terminate. In these lectures, a finite rational expansion does not count as eventually periodic. Source: course book, Section 3.7.

3. The two periodicity theorems

Minutes 3 to 5

State the destination, without trying to prove both results now. Use $\sqrt{2}$ as the simplest example of a quadratic irrational. Today proves the forward direction of Lagrange. Lecture 2 proves the converse and both directions of Galois. Clarify the names gently: Legendre's continued-fraction theorem concerns exceptionally good rational approximations. Lagrange is the eventual-periodicity theorem. The theorem statements use simple continued fractions. Source: course book, Sections 3.7–3.9. External cross-check: Corless, Periodic and Fixed Points of the Gauss Map.

4. A Möbius transformation

Minutes 5 to 7

Use the notation from your preceding lecture, while making clear that $T_M(t)$ is a fractional action rather than ordinary matrix multiplication by a scalar. The denominator condition is essential. With rational coefficients, any finite pole is rational, so an irrational input never hits a pole. Infinity is a convenient optional convention, not a prerequisite for today's proofs. Source: book 3.2.

5. Why a matrix acts on a number

Minutes 7 to 9

This is the projective viewpoint without projective-space terminology. Emphasize the ratio. The columns are ordinary vectors, and the fractional transformation records their coordinate ratio. Ask why M and $-M$ produce the same map. The identity matrix and all its nonzero scalar multiples therefore represent the identity transformation. We will need that observation when excluding degenerate fixed-point equations. Source: book 3.2 and Appendix A.2.

6. Composition

Minutes 9 to 12

Here M is translation by 1 and N is inversion. Compute the product with the class, then compare it with the nested function. If students already remember the general substitution proof, do not repeat its four coefficients. The first backup slide contains that calculation if requested. Ask what happens in the opposite order: $NM = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ acts by $1/(t+1)$. This makes the order of multiplication memorable. Source: book 3.2.

7. An inverse, by solving for t

Minutes 12 to 14

Let students supply the rearrangement $(a - cy)t = dy - b$. An important corollary is that an invertible rational Möbius transformation preserves irrationality: if its output were rational, its inverse would make the input rational. There is no hidden pole in this argument. For $y = T_M(t)$, $a - cy = (ad - bc)/(ct + d) \neq 0$. We use this fact later to rule out a merely linear equation. Source: book 3.2 and the proof in 3.7.

8. One continued-fraction digit

Minutes 14 to 16

The tail t here is a whole remaining continued fraction. The letter D in $D(a)$ means a digit matrix. In Lecture 2, the plain letter D will instead denote a discriminant under a square root; point out the contextual distinction if needed. All finite digit products have determinant ± 1 and integer inverses. This is the exact sense in which we can undo a prefix. Source: book 3.2–3.3.

9. A block of digits

Minutes 16 to 18

Translate the displayed continued fraction back into nested functions. All digits after the integer part are positive. The last entry t is the tail, so it need not be an integer. The example fixes the period matrix that will run through the rest of the lecture. Have students keep $\begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}$ on their paper. Source: book 3.2 and the worked example in 3.7.

10. A repeating tail

Minutes 18 to 20

This is the central idea. Pause before showing the fixed-point equation and ask what remains after removing a complete block. An infinite simple continued fraction has a well-defined irrational limit; this follows from the convergent bounds already developed in class. A backup gives the short irrationality argument. Positivity of the period digits makes B, C strictly positive, even for a one-digit period. Source: book 3.7.

11. A fixed point gives an equation

Minutes 20 to 22

Keep this algebraic step brief. The nontrivial part will be understanding exactly which quadratics repeat. At this point the number is irrational because the continued fraction is infinite, so it is a quadratic irrational, not merely some root of a reducible quadratic. Contrast the period matrix with the identity: $t = T_I(t)$ only yields $0 = 0$. Source: book 3.7.

12. Your turn: the block $1, 2$

Minutes 22 to 25

Give students a real minute of silence, then ask for a quadratic before asking for its roots. Expected equation: $2x^2 - 2x - 1 = 0$. Expected roots: $(1 \pm \sqrt{3})/2$. The continued fraction exceeds 1, so it is the positive root. If someone takes the negative root, ask which interval contains $1 + 1/(2 + 1/(\cdots))$. Advance only after discussion. This is book Exercise 15, expanded into a classroom prompt.

13. The periodic root

Minutes 25 to 27

The prime denotes the other algebraic root, not a derivative. Do not yet explain the significance of the interval $(-1, 0)$. Point out the pattern and promise that Lecture 2 will explain it exactly. One can check the value by substituting it into $(3x + 1)/(2x + 1)$. Source: book 3.7.

14. A finite prefix

Minutes 27 to 29

Emphasize the distinction between a periodic tail and the original number. We know a quadratic equation for the tail x, but must produce one for the number y with its prefix. Let students suggest substitution or inversion. Direct substitution works, but conjugation packages it into one reusable identity. Source: book 3.7, equation (3.3).

15. The prefix conjugation identity

Minutes 29 to 32

Read the composition from right to left: remove the prefix, apply the period to its fixed tail, and put the prefix back. Spend the full three minutes on why each equality holds. For students comfortable with function notation, write MNM^{-1} directly while saying that the matrices act fractionally. An empty prefix means $M = I$. Source: book 3.7, equation (3.3).

16. Why the new equation is genuinely quadratic

Minutes 32 to 35

This is the rigor checkpoint. If $K = \lambda I$, then $N = M^{-1}KM = \lambda I$, contradicting the positive lower-left entry of the period matrix. Thus the three coefficients $c, d - a, -b$ are not all zero. If $c = 0$, a nonzero rational linear polynomial would vanish at the irrational y, which is impossible. Rationality of a Möbius image would also imply rationality of its inverse image. The backup fixed-point lemma gives this argument in isolation. Source: book 3.7.

17. The equation for $\sqrt{3}$

Minutes 35 to 38

Have students check at least one row of the matrix product. For an intermediate check, $MN = \begin{pmatrix} 5 & 2 \\ 3 & 1 \end{pmatrix}$. The positive sign follows from $y = 1 + 1/x > 0$. This example both verifies the familiar continued fraction of $\sqrt{3}$ and shows what conjugation actually does to the quadratic equation. Source: book 3.7 and Exercise 21.

18. Your turn: a different prefix

Minutes 38 to 41

Allow about 90 seconds for the calculation. Students who know $1/x = \sqrt{3} - 1$ can immediately identify $y = 1+\sqrt{3}$, but ask them to check the conjugated matrix too. The result is $\begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix}$, giving $y^2 - 2y - 2 = 0$. The digit stream is $2, 1, 2, 1, \ldots$, so its pure period is $[\overline{2, 1}]$. Source: book Exercise 21.

19. A prefix that disappears

Minutes 41 to 43

Distinguish a particular written decomposition into prefix and period from the shortest possible preperiod of the actual digit sequence. Here the first digit already belongs to a cycle of length two. This is an important example to bring back after Galois's criterion, since the conjugate $1 - \sqrt{3}$ lies in $(-1, 0)$. Source: book 3.9.

20. Quadratic conjugates

Minutes 43 to 45

Assume D is a positive nonsquare integer. Clarify again that the prime is algebraic conjugation, not differentiation. All we need is that changing the sign of $\sqrt{D}$ commutes with adding, multiplying, and taking reciprocals when the denominators are nonzero. The complete quotients are rational transformations of the original surd, so their conjugates follow the same algebra. Source: book 3.7–3.8.

21. Reduced surds

Minutes 45 to 48

Ask which inequality fails in each of the two negative examples. For $\sqrt{2}$ it is the lower bound on the conjugate. For $\sqrt{2}/2$ it is $x > 1$. Galois says that this definition is exactly the right one for pure periodicity. Today students need the definition and examples; Lecture 2 explains the mechanism. Source: book 3.8–3.9.

22. What we have proved

Minutes 48 to 50

Expected exit answer: the identity transformation fixes every real number, so the resulting polynomial is $0 = 0$. We needed a nonscalar transformation, supplied by the positive lower-left entry of the period matrix. Stop here at 50 minutes. The remaining slides are backups, not part of the planned lecture. Suggested preparation: review complete quotients and the fact that convergents p_k/q_k tend to the original irrational.

Optional slides after the 50-minute stopping point

Backup 1. Composition in coordinates

Use only if students request the general composition calculation. The equation holds wherever the displayed finite formulas are defined, and globally if infinity is included in the projective convention. Source: book 3.2.

Backup 2. The nonscalar fixed-point lemma

This separates the elementary lemma from any special continued-fraction properties. Note that being invertible does not by itself exclude the identity transformation. One needs the nonscalar assumption as well as an irrational fixed point. Source: book Lemma 3.12.

Backup 3. Why an infinite simple continued fraction is irrational

The alternating-convergent argument produces a limit for any infinite positive digit sequence, not just for the expansion of a number already known to be irrational. For a rational candidate p/q in lowest terms, increasing denominators ensure that the convergents eventually differ from it. The integer numerator of their difference then has absolute value at least one. A weak upper bound is enough here. Source: book 3.7.

Lecture 2: Lagrange, Galois, and reduced surds

1. Lagrange, Galois, and reduced surds

Minutes 0 to 1

Begin with a quick reminder of the end of Lecture 1: a repeating tail is a fixed point, and attaching a prefix conjugates its matrix. Today's new work proves the converse and the exact pure-periodicity criterion. Students need the complete-quotient algorithm and convergence of p_k/q_k, but no algebraic number theory. Source: book 3.8–3.9.

2. Recall: the complete quotient

Minutes 1 to 3

Rebuild the notation rather than assuming everyone remembers the subscript convention. The integer part of the original number can be zero or negative. Every later complete quotient is greater than 1, since it is the reciprocal of a nonzero fractional part. A repeated complete quotient makes all future digits repeat because the algorithm is deterministic. Source: book 3.5 and 3.8.

3. The target theorem

Minutes 3 to 5

Recall that x' means the other root of the rational quadratic equation, or the value obtained by reversing the sign of its square root. The reducedness conditions are not a conclusion about arbitrary real numbers; they are defined for quadratic irrationals. The key logical distinction is eventual versus pure periodicity. Source: book Theorems 3.13 and 3.17. External cross-check: Corless, Periodic and Fixed Points of the Gauss Map.

4. Conjugates follow the same algebra

Minutes 5 to 7

This is a common source of confusion. Conjugation changes the algebraic number but leaves the rational integer a_n fixed. It does not commute with taking a floor. The inequality uses the floor of the positive complete quotient. Ask students why the reciprocal of a number less than -1 lies in $(-1, 0)$. Once a negative conjugate appears, the next tail is reduced. Source: book 3.8, Step 1.

5. Reduced tails stay reduced

Minutes 7 to 9

The first inequality follows from the fractional-part algorithm. The second follows from the previous slide. Do not skip the last sentence: forward invariance is not the same as eventual entry. This logical separation also prevents a circular proof of Lagrange's theorem. Source: book 3.8, Step 1.

6. Every quadratic irrational reaches a reduced tail

Minutes 9 to 12

Give the full sign argument aloud. Since $p_k/q_k \to x$ and $x' \neq x$, the difference $x' - p_k/q_k$ eventually has the fixed sign of $x' - x$. Multiplication by positive q_k preserves that sign. The two expressions in the ratio therefore have the same sign, and the minus sign makes x'_n negative. Lecture 1's inverse formula gives the displayed identity. A backup derives it in detail. This covers negative original numbers too. Source: book 3.8, Step 1.

7. An integer description of a surd

Minutes 12 to 14

Explain what the state remembers: the fixed square root never changes, so only two integers need to evolve. For completeness, if $Ax^2 + Bx + C = 0$, choose the common sign of the coefficients so that $x = (-B + \sqrt{B^2 - 4AC})/(2A)$. Then $D = B^2 - 4AC$, $P_0 = -B$, $Q_0 = 2A$, and $D - P_0^2 = -4AC$ is divisible by Q_0. Do not automatically insist that the initial denominator be positive. The first backup shows this normalization. Source: book 3.8, Step 2.

8. The exact recurrence

Minutes 14 to 17

Rationalize the denominator explicitly. The sign in $P_{n+1} = a_n Q_n - P_n$ is easy to reverse accidentally, so check it against $x_n - a_n = (\sqrt{D} - P_{n+1})/Q_n$. After rationalization, the numerator has precisely the same form as before. We have not yet proved the new denominator is an integer; that is the next slide. Source: book 3.8, equation (3.4).

9. Why the recurrence stays integral

Minutes 17 to 19

This version uses divisibility directly, so students need not already know congruence notation. Since D is a nonsquare integer, $D - P_{n+1}^2$ is never zero. Thus Q_{n+1} is a nonzero integer. The final equality says that Q_{n+1} itself divides $D - P_{n+1}^2$, completing the induction. Source: book 3.8, Step 2.

10. The bounds from reducedness

Minutes 19 to 22

This is the compactness argument entirely in elementary integer language. The positive difference $x - x'$ first forces $Q > 0$, so it is safe to multiply the conjugate inequalities by Q. The lower bound becomes $P > \sqrt{D} - Q > -\sqrt{D}$ because $Q < 2\sqrt{D}$. No uniform bound on the period length is needed. Source: book 3.8, equation (3.5).

11. Finitely many states

Minutes 22 to 24

Name the pigeonhole principle. Repetition of a pair means repetition of the complete quotient, and deterministic updating reproduces the entire future, not just one digit. We have now proved both directions of Lagrange using fixed points, convergence, rationalization, and finite integer bounds. We still have not removed an initial segment for a reduced starting state. Source: book 3.8.

12. The cycle for $\sqrt{3}$

Minutes 24 to 26

Do one transition explicitly: from $(P, Q, a) = (0, 1, 1)$ we get $P_{\text{new}} = 1$, $Q_{\text{new}} = 2$. From $(1, 2, 1)$ we get $(1, 1)$. The equality of states at $n = 1$ and $n = 3$ certifies period two. The finite table illustrates the proof, but the bounds explain why every quadratic irrational must eventually produce such a repetition. Source: book Example 3.16.

13. Your turn: $\sqrt{6}$

Minutes 26 to 29

Give about two minutes of working time and one minute to compare. The first digit is 2. The states are $(0, 1)$, $(2, 2)$, $(2, 1)$, $(2, 2)$, with digits $2, 2, 4, 2$. If a student gets a negative denominator after the first step, check the sign of P_{n+1}. The interval $2 < \sqrt{6} < 3$ is enough to determine all digits in this example without a decimal calculator. This is a new worked classroom example using book (3.4).

14. The period of $\sqrt{6}$

Minutes 29 to 30

Keep the answer reveal brief. The first two reduced complete quotients are $(2 + \sqrt{6})/2$ and $2 + \sqrt{6}$. Their conjugates are $(2 - \sqrt{6})/2$ and $2 - \sqrt{6}$, both in $(-1, 0)$. This example gives a quick independent check that the recurrence and reducedness conditions are working together.

15. Finiteness allows an initial segment

Minutes 30 to 32

This is the conceptual turning point of the second lecture. The reduced states form a finite forward-invariant set, but that alone would not prove that an orbit starts on its cycle. The little finite-map example exhibits the missing possibility. Galois's criterion succeeds because the continued-fraction step has unique reduced predecessors. Source: book 3.9.

16. Galois's criterion

Minutes 32 to 34

Recall the definition from Lecture 1. We already know that every reduced orbit stays reduced and eventually repeats. The next two slides strengthen that conclusion to pure periodicity. Afterward we prove necessity. Keep the logical direction explicit throughout. Source: book Theorem 3.17.

17. A successor remembers its reduced predecessor

Minutes 34 to 37

The strict inequalities use precisely $-1 < u' < 0$. This is why the definition of reducedness is so well chosen. Stress that the assertion concerns reduced predecessors. Without that restriction, many different integers could be placed before the same tail. For a quick check, take $z = (1 + \sqrt{3})/2$. Then $-1/z' = 1 + \sqrt{3}$, so $a = 2$ and the predecessor is $1 + \sqrt{3}$. Source: book 3.9, equation (3.6).

18. Rewinding a repeated state

Minutes 37 to 39

If $i = 0$, the conclusion already holds. Otherwise apply the predecessor formula repeatedly until reaching index zero. Equal quadratic numbers have equal conjugates, so the recovered digit and predecessor agree. Now deterministic forward updating gives pure periodicity. The point is not that we ran the ordinary floor algorithm backward on arbitrary reals, but that reduced surds carry enough information in their conjugates. Source: book 3.9.

19. Why a pure period must be reduced

Minutes 39 to 41

This is the other implication of Galois. There is no circular appeal to Galois in the eventual-entry argument: it used convergence of the ordinary convergents and the inverse prefix matrix. Together with the preceding two slides, this completes the characterization. Source: book 3.9.

20. A reducedness test

Minutes 41 to 44

Allow about 90 seconds, then ask for reasons rather than just yes/no answers. The middle two are reduced. For $\sqrt{2}$ the conjugate is below -1. For $\sqrt{2}/2$ the original number is below 1. All four are quadratic irrationals, so all four are eventually periodic by Lagrange. Do not reveal the next slide until the discussion finishes. Source: book 3.9 and Exercise 23.

21. The test, checked

Minutes 44 to 46

For the golden ratio, the inequalities follow from $2 < \sqrt{5} < 3$. Its period consists of the digit 1. For $1 + \sqrt{3}$, the period is $2, 1$. Ask a student to distinguish the roles of the two theorems in the final line. Source: book 3.9.

22. A visible prefix can belong to the period

Minutes 46 to 47

Return to the example from Lecture 1. The earlier calculation established the value. Galois now explains the classification. A written bar need not mark the earliest possible start of repetition. Source: book 3.9.

23. Lagrange and Galois

Minutes 47 to 49

Keep the quadratic-irrational hypothesis on the Galois line explicit orally. The two lectures together used only matrix algebra, convergence of convergents, rationalization, inequalities, and the pigeonhole principle. Mention optional links to Pell's equation and reversal of the period only if time remains. Both appear in the backup slides. Source: book 3.7–3.10.

24. Exit question

Minutes 49 to 50

Expected answer: finiteness forces some repeated state and therefore eventual periodicity. Unique reduced predecessors let us rewind that repetition to index zero, which gives pure periodicity. Without reversibility, an orbit can have a transient initial segment, as in $0, 1, 1, \ldots$. This is the 50-minute stopping point. Suggested follow-up: book Exercises 24–26, or derive the period of $\sqrt{13}$ using the exact recurrence.

Optional slides after the 50-minute stopping point

Backup 1. Choosing the initial integer state

The discriminant is nonsquare because x is irrational. If the desired root initially has the minus sign before the square root, replacing (A, B, C) by $(-A, -B, -C)$ puts that root in the displayed plus-sign form. For example, $x = -\sqrt{2}$ can be represented with $D = 8$, $P_0 = 0$, $Q_0 = -2$. This explains why a proof for all real quadratic irrationals must allow initial negative denominators. Source: book 3.8, Step 2.

Backup 2. The inverse-prefix formula

Use this if the sign argument in main slide 6 needs more time. The denominators are nonzero because the conjugate is irrational and the convergents are rational. For $n \geq 2$, the relevant denominators q_{n-2} and q_{n-1} are positive. Convergence toward x, not any assumption of periodicity, gives the common sign. Source: book 3.8, Step 1.

Backup 3. The conjugate reverses the period

Both B and C are positive. The transformed quadratic therefore has one positive root and one negative root. By Galois, $-1/x' > 0$; the reversed periodic continued fraction is positive too, so they are the same root. For a concrete check, $[\overline{1, 2, 3}] = (4 + \sqrt{37})/7$, while the reversed period is $(4 + \sqrt{37})/3$. Source: book Corollary 3.19 and Exercise 26.

Backup 4. The same matrix produces Pell solutions

This is an optional connection, not an additional theorem needed for either periodicity proof. The determinant-one matrix acts both as a Möbius transformation fixing $\sqrt{3}$ and as an ordinary linear transformation preserving the quadratic form $p^2 - 3q^2$. Avoid claiming that this calculation alone proves the classification of all Pell solutions. Source: book 3.10 and Exercise 18.

Sources and notation

The primary source for these lectures is the expanded course book: Chapter 3, especially Sections 3.2 and 3.7–3.10, together with Appendix A. The printed book numbers its three central identities as (3.3), prefix conjugation; (3.4), the quadratic-state recurrence; and (3.6), the unique reduced predecessor.

The notation $T_M(t)$ means the fractional action of a matrix, not ordinary multiplication by a scalar. The prime on a quadratic irrational denotes its algebraic conjugate, not a derivative. Every continued fraction in the lectures is simple, with positive digits after its integer part.

Robert M. Corless, *Continued Fractions and Chaos*, Periodic and Fixed Points of the Gauss Map, states the Lagrange and Galois characterizations. Giovanni Panti, A general Lagrange Theorem (2007), gives a broader setting for periodicity results beyond ordinary simple continued fractions. The lectures need none of those generalizations.

For a presentation setup reference, Zoom documents sharing a slide-show window. The lecture PDFs themselves do not depend on Zoom.